

## JEJU INTL, Korea, Republic of

WMO: 471820

Lat: 33.511N

Lon: 126.493E

Elev: 36

StdP: 100.89

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -0.1       | 0.9 | -8.2                          | 1.9 | 3.5  | -7.1 | 2.1 | 3.6  | 13.4                  | 3.5  | 12.3 | 3.3  | 6.6                   | 350  | 0.462 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 5.1                    | 32.1            | 25.9 | 31.0 | 26.1 | 30.0 | 25.7 | 27.9                | 29.9 | 27.0 | 29.1 | 26.2 | 28.6 | 6.2                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 27.2                            | 23.1 | 29.3 | 26.5 | 22.2 | 28.7 | 25.8 | 21.2 | 28.1 | 90.3          | 30.0 | 85.7 | 28.8 | 82.2 | 28.3 | 31.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 12.2              | 10.8 | 9.6 | -2.0                       | 34.4 | 1.7                | 1.1 | -3.2                                               | 35.2 | -4.2       | 35.9 | -5.1       | 36.5 | -6.4       | 37.3 |    |    |
|                   |      |     | -3.5                       | 28.1 | 1.4                | 1.5 | -4.5                                               | 29.2 | -5.4       | 30.0 | -6.1       | 30.9 | -7.1       | 32.0 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 15.8   | 5.6   | 6.3   | 9.1   | 13.4  | 17.5  | 21.2  | 26.0  | 27.0  | 23.0  | 18.4  | 13.0  | 7.9  |
|                                                                                  | DBStd       | 7.76   | 2.88  | 3.18  | 3.12  | 2.91  | 2.37  | 2.12  | 2.57  | 2.06  | 2.07  | 2.48  | 3.16  | 3.02 |
|                                                                                  | HDD10.0     | 397    | 141   | 112   | 54    | 4     | 0     | 0     | 0     | 0     | 0     | 8     | 79    |      |
|                                                                                  | HDD18.3     | 1728   | 396   | 338   | 286   | 151   | 43    | 2     | 0     | 0     | 29    | 161   | 322   |      |
|                                                                                  | CDD10.0     | 2499   | 3     | 7     | 27    | 106   | 234   | 337   | 496   | 526   | 390   | 260   | 99    | 14   |
|                                                                                  | CDD18.3     | 788    | 0     | 0     | 0     | 2     | 19    | 89    | 238   | 268   | 140   | 31    | 2     | 0    |
|                                                                                  | CDH23.3     | 5893   | 0     | 0     | 1     | 9     | 56    | 275   | 2162  | 2726  | 622   | 41    | 1     | 0    |
|                                                                                  | CDH26.7     | 1685   | 0     | 0     | 0     | 1     | 7     | 38    | 682   | 879   | 76    | 3     | 0     | 0    |
| Wind                                                                             | WSAvg       | 4.1    | 4.7   | 4.4   | 4.4   | 4.1   | 3.7   | 3.8   | 4.0   | 3.9   | 4.1   | 4.0   | 4.1   | 4.7  |
| Precipitation                                                                    | PrecAvg     | 1466   | 65    | 64    | 79    | 95    | 94    | 176   | 214   | 258   | 209   | 83    | 76    | 54   |
|                                                                                  | PrecMax     | 2526   | 170   | 164   | 228   | 211   | 284   | 694   | 706   | 663   | 940   | 421   | 231   | 131  |
|                                                                                  | PrecMin     | 773    | 12    | 8     | 20    | 15    | 13    | 12    | 15    | 53    | 8     | 5     | 2     | 11   |
|                                                                                  | PrecStd     | 388    | 37    | 36    | 47    | 46    | 52    | 111   | 149   | 153   | 185   | 83    | 49    | 29   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 16.2  | 19.1  | 20.8  | 24.9  | 27.2  | 29.8  | 34.0  | 33.9  | 30.2  | 26.2  | 22.9  | 18.1 |
|                                                                                  |             | MCWB   | 13.3  | 15.2  | 14.9  | 16.4  | 18.9  | 24.1  | 25.5  | 26.3  | 25.3  | 22.0  | 18.0  | 14.6 |
|                                                                                  | 2%          | DB     | 13.2  | 15.9  | 18.1  | 21.9  | 24.8  | 27.8  | 32.1  | 32.2  | 28.2  | 24.2  | 21.0  | 15.9 |
|                                                                                  |             | MCWB   | 10.1  | 13.1  | 13.6  | 16.9  | 18.9  | 23.9  | 25.7  | 26.2  | 24.6  | 20.7  | 17.6  | 12.8 |
|                                                                                  | 5%          | DB     | 11.8  | 13.0  | 16.1  | 20.0  | 23.2  | 26.1  | 31.0  | 31.1  | 27.2  | 23.1  | 19.2  | 14.1 |
|                                                                                  |             | MCWB   | 8.9   | 10.0  | 12.5  | 15.6  | 17.8  | 22.4  | 26.0  | 26.3  | 23.9  | 19.6  | 16.1  | 11.2 |
|                                                                                  | 10%         | DB     | 10.1  | 11.1  | 14.2  | 18.2  | 22.0  | 24.9  | 29.9  | 30.2  | 26.1  | 22.1  | 18.0  | 12.9 |
|                                                                                  |             | MCWB   | 7.6   | 8.2   | 11.0  | 14.5  | 17.3  | 21.7  | 25.6  | 26.0  | 23.0  | 18.5  | 15.1  | 9.9  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 14.2  | 15.7  | 16.6  | 19.7  | 23.0  | 26.2  | 29.0  | 29.2  | 27.1  | 23.6  | 19.8  | 15.7 |
|                                                                                  |             | MCDB   | 15.7  | 18.3  | 18.9  | 21.8  | 25.4  | 27.4  | 31.1  | 31.4  | 28.5  | 25.4  | 21.7  | 17.7 |
|                                                                                  | 2%          | WB     | 10.9  | 13.3  | 14.7  | 18.2  | 21.1  | 24.7  | 27.8  | 28.3  | 25.8  | 21.8  | 18.2  | 13.2 |
|                                                                                  |             | MCDB   | 12.7  | 15.4  | 17.4  | 20.9  | 23.1  | 26.7  | 29.7  | 30.3  | 27.5  | 23.4  | 20.2  | 15.2 |
|                                                                                  | 5%          | WB     | 9.3   | 10.7  | 13.1  | 16.6  | 19.4  | 23.6  | 26.9  | 27.3  | 24.7  | 20.6  | 16.8  | 11.5 |
|                                                                                  |             | MCDB   | 11.2  | 12.5  | 15.7  | 18.7  | 21.7  | 25.7  | 29.0  | 29.5  | 26.6  | 22.4  | 18.9  | 13.8 |
|                                                                                  | 10%         | WB     | 7.8   | 8.8   | 11.5  | 15.3  | 18.1  | 22.4  | 26.1  | 26.5  | 23.7  | 19.3  | 15.4  | 10.2 |
|                                                                                  |             | MCDB   | 9.8   | 10.9  | 13.8  | 17.6  | 20.7  | 24.2  | 28.5  | 28.9  | 25.7  | 21.7  | 17.7  | 12.5 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 4.5   | 5.5   | 6.5   | 7.1   | 7.0   | 5.6   | 5.3   | 5.1   | 4.8   | 5.3   | 5.5   | 4.9  |
|                                                                                  |             | MCDBR  | 7.6   | 8.9   | 10.0  | 9.7   | 9.1   | 7.2   | 6.8   | 6.3   | 6.0   | 6.3   | 7.4   | 7.4  |
|                                                                                  |             | MCWBR  | 6.9   | 7.7   | 8.3   | 7.2   | 6.4   | 4.9   | 3.6   | 3.5   | 4.2   | 5.1   | 6.4   | 6.8  |
|                                                                                  | 5% WB       | MCDBR  | 6.9   | 8.5   | 9.2   | 8.2   | 8.3   | 6.3   | 6.1   | 5.7   | 5.5   | 5.7   | 6.6   | 7.0  |
|                                                                                  |             | MCWBR  | 6.8   | 7.9   | 8.1   | 6.8   | 6.7   | 5.1   | 4.2   | 3.9   | 4.3   | 5.2   | 6.3   | 6.9  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.461  | 0.501 | 0.553 | 0.562 | 0.571 | 0.603 | 0.583 | 0.539 | 0.496 | 0.467 | 0.460 | 0.454 |      |
|                                                                                  | taud        | 1.958  | 1.893 | 1.786 | 1.818 | 1.852 | 1.865 | 1.965 | 2.098 | 2.153 | 2.097 | 2.037 | 1.980 |      |
|                                                                                  | Ebn at Noon | 711    | 729   | 728   | 747   | 746   | 719   | 731   | 758   | 772   | 756   | 709   | 688   |      |
|                                                                                  | Edh at Noon | 147    | 174   | 209   | 212   | 207   | 203   | 183   | 158   | 143   | 140   | 135   | 137   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.08   | 2.91  | 4.11  | 4.90  | 5.24  | 4.60  | 4.91  | 5.03  | 4.19  | 3.69  | 2.58  | 1.93  |      |
|                                                                                  | RadStd      | 0.23   | 0.34  | 0.33  | 0.40  | 0.56  | 0.55  | 0.68  | 0.51  | 0.40  | 0.37  | 0.28  | 0.17  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | N/A   | N/A     | -2.04  | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (14 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +64     | N/A     |

Nomenclature: See separate page